

Francesco Bronzino

École Normale Supérieure de Lyon, 46 allée d'Italie 69364 Lyon, France
francesco.bronzino@ens-lyon.fr (+33) 0761063734 www.fbronzino.com

EDUCATION

École Normale Supérieure de Lyon Habilitation à Diriger des Recherches Thesis title: "Bridging the Gap Between Machine Learning and Networked Systems" Committee: I. Chrisment, M. Crovella, M. Dias de Amorim, A. Duda, M. Fiore, I. Guérin-Lassous	2024
Rutgers University Ph.D. in Electrical and Computer Engineering (GPA 3.9/4) Thesis title: "Named-Object Based Services in the Future Internet Architecture" Advisor: Prof. Dipankar Raychaudhuri	2012 - 2016
Università degli Studi di Torino Laurea Magistrale in Computer Science (110/110 summa cum laude) Advisors: Prof. Marco Grangetto and Prof. Giovanni Pau (UCLA)	2009 - 2012
Università degli Studi di Torino Laurea in Computer Science (107/110) Advisor: Prof. Matteo Sereno	2006 - 2009

RESEARCH TOPICS

Data driven network management. Design of systems and data driven models to advance the operationalization of machine learning for network management tasks.

Edge assisted network services. Design and analysis of solutions to support the distributed execution of network based services, *e.g.*, video analytics on heterogeneous edge computing resources.

Privacy oriented network systems. Performance analysis and design of network systems that preserve user privacy, *e.g.*, multi-party relay architectures.

EXPERIENCE

École Normale Supérieure de Lyon Maître de conférences (Associate Professor). Computer Science department and LIP (UMR 5668)	2022 - Now
Institut Universitaire de France (IUF) Junior Fellow.	2025 - Now
NetMicroscope Inc. (University Spin-off) Co-founder and CTO.	2021 - 2025
Université Savoie Mont Blanc Maître de conférences (Associate Professor). IAE Annecy and LISTIC	2020 - 2022
Nokia Bell Labs France Research Scientist. AAAID Lab.	2018 - 2020
Inria, Paris, MiMove Research Group Post-Doctoral Research Fellow. Mentor: Dr. Renata Cruz Teixeira.	2016 - 2018
WINLAB, Rutgers University, ECE Department Graduate Research Assistant. Advisor: Prof. Dipankar Raychaudhuri.	2012 - 2016

SELECTED PUBLICATIONS

Less is More: Optimizing Probe Selection Using Shared Latency Anomalies

T. Sharma, A. Chu, P. Schmitt, F. Bronzino, N. Feamster, and N. Marwell
In ACM CoNEXT 2026, Utrecht, Netherlands.

NetSSM: Multi-Flow and State-Aware Network Trace Generation using State-Space Models

A. Chu, X. Jiang, S. Liu, A. Bhagoji, F. Bronzino, P. Schmitt, and N. Feamster.
In ACM CoNEXT 2026, Utrecht, Netherlands.

CATO: End-to-end Optimization of ML Traffic Analysis Pipelines

G. Wan, S. Liu, F. Bronzino, N. Feamster, and Z. Durumeric.

In USENIX NSDI 2025, Philadelphia, USA.

JITI: Dynamic Model Serving for Just-in-Time Traffic Inference

X. Jiang, S. Liu, S. Naama, [F. Bronzino](#), P. Schmitt, and N. Feamster.
In ACM CoNEXT 2025, Hong Kong, China.

Beyond Data Points: Regionalizing Crowdsourced Latency Measurements

T. Sharma, P. Schmitt, [F. Bronzino](#), N. Feamster, and N. Marwell.
In ACM Sigmetrics 2025, Stony Brook, USA.

NetDiffusion: Network Data Augmentation Through Protocol-Constrained Traffic Generation

X. Jiang, S. Liu, A. Gember-Jacobson, A. Bhagoji, P. Schmitt, [F. Bronzino](#), and N. Feamster.
In ACM Sigmetrics 2024, Venezia, Italy.

VideoJam: Self-Balancing Architecture for Live Video Analytics

Y. Faye, F. Faticanti, S. Jain, and [F. Bronzino](#).
In ACM/IEEE SEC 2024, Rome, Italy.

OVIDA: Orchestrator for Video Analytics on Disaggregated Architecture

M. Singh, P. Rachuri, B. Cao, A. Sharma, V. Bhumireddy, [F. Bronzino](#), S. Das, A. Gandhi, S. Jain.
In ACM/IEEE SEC 2024, Rome, Italy.

LEAF: Navigating Concept Drift in Cellular Networks

S. Liu, [F. Bronzino](#), P. Schmitt, A. Nitin Bhagoji, N. Feamster, H. Garcia Crespo, T. Coyle, and B. Ward.
In ACM CoNEXT 2023, Paris, France.

Traffic Refinery: Cost-Aware Data Representation for Machine Learning on Network Traffic

[F. Bronzino](#), P. Schmitt, S. Ayoubi, H. Kim, R. Teixeira, N. Feamster.
In ACM Sigmetrics 2022, Mumbai, India.

Inferring Streaming Video Quality from Encrypted Traffic: Practical Models and Deployment Experience.

[F. Bronzino](#), P. Schmitt, S. Ayoubi, G. Martins, R. Teixeira, N. Feamster.
In ACM Sigmetrics 2020, Boston, USA.

NOVN: A Named-Object Based Virtual Network Architecture to Support Advanced Mobile Edge Computing Services.

[F. Bronzino](#), S. Maheshwari, D. Raychaudhuri, I. Seskar.
Elsevier Pervasive and Mobile Computing Journal 69 (2020) 101261.

Scalability and Performance Evaluation of Edge Cloud Systems for Latency Constrained Applications.

S. Maheshwari, D. Raychaudhuri, I. Seskar, [F. Bronzino](#).
In ACM/IEEE SEC 2018, Bellvue, USA.

FUNDING AS PRINCIPAL INVESTIGATOR

Comcast Faculty Innovation Fund Award

Project developing a passive measurement platform for home network performance analysis. **50k\$, 2026-2027**

IUF ORGANIC: Observing Application Quality at Network Scale

Project developing a new paradigm for taking advantage of modern quality inference models to allow operators to manage networks at-scale. **75k€ and 2/3 of teaching duties reduction, 2025-2030**

ANR GTTP: Global Tracking Transfer Protocol

Project developing a unified privacy-first geolocalization infrastructure. With Inria and Sentiens. **233k€ (out of project total 501k€), 2025-2028**

CNRS IEA GATEAU: Generative AI Techniques for Network Management

Project developing AI-driven generators for network traces. With University of Chicago . **5k€, 2025-2026**

ANR-NSF MINT: Modeling Modern Network Traffic: From Data Representation to Automated Machine Learning

Project studying new methods to represent and model traffic for machine learning driven network management and inference. With University of Chicago and Stanford University. **219k€ (out of project total 602k€), 2021-2025**

ANR PARFAIT: Planning and Learning for AI-Edge Computing

Project studying distributed allocation schemes based on Markov decision processes and Reinforcement Learning for edge computing infrastructures tailored for AI-intensive processing tasks. With Avignon Université, Inria, and CNAM. **135k€ (out of project total 559k€), 2022-2025**

FIL INTERFERE: Stressing Systems Security Through On-the-fly Network Traffic Generation

Project studying the generation of real-time network traffic to create auditing tools for stressing systems and their security. With INSA Lyon **10k€, 2023-2025**

AAP USMB: Decentralized Live Video Analytics at the Network Edge

Project studying how to integrate mobile video sources into the video analytics processing infrastructure using heterogeneous compute resources at the edge of the network. **3k€, 2022**

FUNDING AS PROJECT MEMBER**ANR NEMIOT: Planning and Learning for AI-Edge Computing**

Project studying the network behavior of IoT devices. With Inria and École Polytechnique. **172k€ (out of project total 372k€), 2024-2027**

Four grants through France And Chicago Collaborating in The Sciences (FACCTS)

With the University of Chicago. Four separate grants of **\$22.5k, \$15k, \$15k, and \$15k, 2021-2024**

Industrial contract with CELESTE

Project studying how to automatically extract network configurations satisfying client constraints using machine learning solutions that learn from past configurations rules and apply them to future cases. **Half a PhD thesis funding, 84k€, 2021-2024**

TEACHING**École Normale Supérieure de Lyon**

Networks, Network measurements, Performance Evaluation, and Architectures and Operating Systems **2022 - Now**

IAE, Université Savoie Mont Blanc

Intro to programming, Machine learning, Computer networks, and Operating systems **2020 - 2022**

WINLAB, Rutgers University, ECE Department

Communication Networks II and Introduction to Computers for Engineers **2013 - 2016**

ADVISING**Ph.D. Students**

- Augustin Laouar **2025 - Now**
"Global Tracking Transfer Protocol". With L. Desgeorges (U Lyon).
- Saloua Naama **2021 - Now**
"New Network Systems Mechanisms to Support Massive Machine Learning". With Prof. Salamatian (USMB).
- Johann Hugon **2022 - 2025**
"Modeling Modern Network Traffic for Automated Machine Learning in Network Systems". With Prof. Busson (U Lyon).
- Youssouph Faye **2022 - 2025**
"Distributed Edge Cloud Architecture for Executing AI-Based Applications".
- Razanne Abu-Aisheh **2020 - 2023**
"Context-Aware Information Gathering and Processing Towards Supporting Autonomous Systems in Industry 4.0 Scenarios". With Dr. Thomas Watteyne (Inria) and Dr. Myriana Rifai (Nokia Bell Labs).
First job: Research associate at Bristol University.

Postdocs

- Adrien Schoen **2025 - Now**
"AI for network management".
- Francescomaria Faticanti **2023 - 2025**
"Network inference at the edge". First job: Inria SFP.

Interns

- Alain Gérard Mbo, Syed Muhammad Ali Shah, Alyce Théobald, Gaëlle Manuela Yonga Yonga, Alexis Carré, Ciro Guida, Jules Marmier, Malo Jaffré, Gaëtan Nodet, Duncan Bordin, Hakim Ouhida, Setareh Nouri, Ayoub Ben Ameur.

AWARDS AND HONORS

- Comcast Faculty Innovation Award 2026.
- Junior fellow of the Institut universitaire de France (IUF) 2025-2030.
- Recipient of the GDR RSD Junior Researcher Award 2025.
- Recipient of the George Shultz Innovation Fund (\$200k Startup Fund)
- Wall Street Journal Front Page Feature (August 19, 2019, "The Truth About Faster Internet: It's Not Worth It").
- 2016/2017 recipient of the "ECE Graduate Program Academic Achievement Award" from the ECE department at Rutgers University.
- Recipient of the best paper award at the RAFNET 2017 workshop.
- Recipient of the best paper award at the ACM 2012 MobiArch workshop.

PRESS

- "Rethinking Geolocation on the Internet", Internet Society Pulse.
- "Enhancing Internet Resilience with Spatial Analysis", Internet Society Pulse.
- "The Truth About Faster Internet: It's Not Worth It". The Wall Street Journal, Aug 21st 2019 front page print and online editions.
- "SES and Rutgers University test satellite content delivery network for streaming, OTT, and 5G". SES Blog.
- "WINLAB develops infrastructure for potential new internet", The Daily Targum, Nov 2nd 2016

PROFESSIONAL SERVICE

Technical Program Committee Member: ACM SIGCOMM 2026, ACM CoNEXT 2026, ACM IMC 2026, ACM Sigmetrics 2026, ACM HotNets 2025, ACM CoNEXT 2025, ACM Sigmetrics 2025, ACM IMC 2025, USENIX NSDI 2025, IEEE Infocom 2025, ACM IMC 2024, ACM CoNEXT 2024, IEEE Infocom 2024 (Distinguished Member), IEEE MedComNet 2024, CoRes 2024, ACM Sigmetrics 2023 (best paper award committee), IEEE Infocom 2023 (Distinguished Member), CoRes 2023, IEEE MedComNet 2023, IEEE Smartcomp 2023, IEEE CloudCom 2023, IEEE Infocom 2022 (Distinguished Member), IEEE MedComNet 2022, IEEE Smartcomp 2022, IEEE MedComNet 2021, ACM GoodIT 2021, ITC 33, IEEE Infocom 2021, IEEE Infocom 2020, ACM FRUGALTHINGS 2020 Workshop, IEEE Infocom 2019 (Distinguished Member), IEEE VTC2018-Fall, ACM NEAT 2018 Workshop, ACM MMSys 2018, ACM MobiMWareHN 2017 Workshop, ACM S3 2016 Workshop.

Organization Committee Member: IEEE CloudCom 2023, IEEE SMARTCOMP 2020, IEEE CCNC 2018, IEEE CCNC 2017, ACM MobiMWareHN 2017 Workshop, IEEE CCNC 2016, ACM MobiCom 2016, ACM MobiSys 2015.

Editor: Editor for Elsevier Computer Networks (from 2026), Associate Editor for Elsevier Computer Communications (2021-2026).

Reviewer: IEEE TMC, IEEE TSC, ACM TOMM, Elsevier Computer Networks, etc.

Proposal reviewer: Reviewer for grant proposals for the ANR 2024 ASTRID program. Reviewer for grant proposals for the Swiss National Science Foundation (2018).

ACADEMIC SERVICE

Academic duties

Coordinator for the Networking Reading Group at ENS Lyon. 2022 - 2026

Correspondent for for the CS department for the green transition at ENS Lyon. 2023 - Now

Head of international relations for the TM department at IAE Savoie Mont Blanc. 2021 - 2022

Animation committee and communication committee at LISTIC Laboratory. 2021 - 2022

Hiring committee member

Maître de conférences in computer science at ENS de Lyon / LIP. 2026

Maître de conférences in computer science at Université Grenoble Alpes / LIG. 2026

CDD Faculty position in computer science at INSA Lyon / CITI. 2026

Maître de conférences in computer science at Sorbonne Université / LIP6. 2025

Enseignants contractuel LRU, at Université d'Avignon / LIA. 2025

Ph.D. jury member

Ph.D. proposal and defense of Taveesh Sharma, University of Chicago. 2026

Ph.D. proposal and defense of Xi Jiang, University of Chicago. 2026

Ph.D. defense of Shurok Khozam, Telecom SudParis. 2026

Ph.D. thesis reviewer and defense of Omar Anser, Inria Nancy. 2026

Ph.D. proposal and defense of Sri Pramodh Rachuri, Stony Brook University. 2026

Ph.D. defense of Chao Wang, EURECOM. 2025

Ph.D. CSI of Hugo Rimmlinger, Sorbonne Université. 2024-2025

Ph.D. defense of Davide Aguiari, Università degli Studi di Cagliari and Sorbonne Université. 2021

MAIN REFERENCES

Dipankar Raychaudhuri, Distinguished Professor, Rutgers University, ray@winlab.rutgers.edu

Renata Cruz Teixeira, Senior Research Scientist at Netflix and Inria, rteixeira@netflix.com

Nick Feamster, Neubauer Professor, University of Chicago, feamster@uchicago.edu